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Solution

Speed of the cyclist, $v = 20\ m\ s^{-1}$

Angle of bending with vertical, $\theta = 30^{o}$

Equation for angle of bending, $\tan\theta = \frac{v^2}{rg}$

Rewriting the above equation for radius $r = \frac{v^2}{\tan\theta\, g}$

Substituting,

$$r = \frac{(20)^2}{(\tan 30^{\circ})\times 10} = \frac{20\times 20}{(\tan 30^{\circ})\times 10}$$

$$= \frac{400}{\left(\frac{1}{\sqrt{3}}\right)\times 10}$$

$$r = (\sqrt{3})\times 40 = 1.732\times 40$$

$$r = 69.28\ m$$

5.4 MOMENT OF INERTIA

In the expressions for torque and angular momentum for rigid bodies (which are considered as bulk objects), we have come across a term $\sum m_i r_i^2$. This quantity is called moment of inertia (I) of the bulk object. For point mass m_i at a distance r_i from the fixed axis, the moment of inertia is given as, $m_i r_i^2$.

Moment of inertia for point mass,

$$I = m_i r_i^2 \qquad (5.37)$$

Moment of inertia for bulk object,

$$I = \sum m_i r_i^2 \qquad (5.38)$$

In translational motion, mass is a measure of inertia; in the same way, for rotational motion, moment of inertia is a measure of rotational inertia. The unit of moment of inertia is, kg m^2. Its dimension is ML^2. In general, mass is an invariable quantity of matter (except for motion comparable to that of light). But, the moment of inertia of a body is not an invariable quantity. It depends not only on the mass of the body, but also on the way the mass is distributed around the axis of rotation.

To find the moment of inertia of a uniformly distributed mass; we have to consider an infinitesimally small mass (dm) as a point mass and take its position (r) with respect to an axis. The moment of inertia of this point mass can now be written as,

$$dI = (dm)r^2 \qquad (5.39)$$

We get the moment of inertia of the entire bulk object by integrating the above expression.

$$I = \int dI = \int (dm)r^2$$

$$I = \int r^2 dm \qquad (5.40)$$

We can use the above expression for determining the moment of inertia of some of the common bulk objects of interest like rod, ring, disc, sphere etc.

5.4.1 Moment of Inertia of a Uniform Rod

Let us consider a uniform rod of mass (M) and length (ℓ) as shown in Figure 5.21. Let us find an expression for moment of inertia of this rod about an axis that passes through the center of mass and perpendicular to